Codebook / Data Dictionary: blue_light_sleep_sim_psqi.csv

Simulated dataset reflecting associations between bedtime blue-light device use and sleep outcomes (Pittsburgh Sleep Quality Index; PSQI) in a university student sample.

Design: Cross-sectional, N = 286 (one row per participant).

General rule: Unless noted, higher values indicate more of the construct; direction (better/worse) is indicated below.

Variable	Type	Units / Range	Definition	Levels / Coding	Direction / Notes
participant_id	integer	1–286	Unique participant identifier	—	—
sex	factor	—	Self-reported sex	female, male	—
age	integer	years (≈18–30)	Age at survey	whole years	—
field_study	factor	—	Academic field of study	medicine, pharmacy	—
owns_device	integer (0/1)	0–1	Owns a smartphone/device	1 = yes, 0 = no (simulated all 1)	—
bedtime_device_use	integer (0/1)	0–1	Any device use at bedtime	1 = yes, 0 = no	Higher = more risk
device_use_minutes_bedtime	numeric	minutes (0–240)	Minutes of device use in bed before sleep	decimal minutes	Higher = more risk
brightness_adjustment	factor	—	Night mode / blue-light filter or similar	on, off	“on” generally protective
room_lights_off	integer (0/1)	0–1	Room lights off while using device in bed	1 = lights off, 0 = lights on	Contextual
device_switched_off	integer (0/1)	0–1	Device switched off before sleep	1 = switched off, 0 = kept on	Higher = better
device_under_pillow	integer (0/1)	0–1	Sleeps with device under pillow	1 = yes, 0 = no	Higher = more risk
sleep_interrupt_check	integer (0/1)	0–1	Checks device if sleep is interrupted	1 = yes, 0 = no	Higher = more risk
use_reason	factor	—	Primary reason for bedtime device use	leisure, reading	reading somewhat protective
sleep_duration_hours	numeric	hours (≈3.5–10)	Estimated nocturnal sleep duration	decimal hours	Higher = better
sleep_efficiency_pct	numeric	% (≈40–99)	Sleep efficiency (time asleep / time in bed)	decimal percent	Higher = better
sleep_latency_min	integer	minutes (0–120)	Time to fall asleep	whole minutes	Higher = worse
sleep_quality_psqi	integer	0–21	Pittsburgh Sleep Quality Index total	>5 indicates poor sleep	Higher = worse
psqi_poor	integer (0/1)	0–1	Poor sleeper per PSQI (>5)	1 = poor sleeper, 0 = good	Higher = worse
fatigue_on_waking_perweek	integer	days/week (0–7)	Days per week waking with fatigue	whole days	Higher = worse
headaches_perweek	integer	days/week (0–7)	Headache frequency	whole days	Higher = worse
irritability_perweek	integer	days/week (0–7)	Irritability frequency	whole days	Higher = worse
caffeine_cups_day	integer	cups/day (0–7)	Daily caffeine consumption	whole cups	Higher = more caffeine
exercise_days_week	integer	days/week (0–7)	Exercise frequency	whole days	Higher = more exercise
chronotype	factor	—	Morningness–eveningness type	morning, intermediate, evening	—
bmi	numeric	kg/m^2 (≈16–38)	Body mass index	one decimal place	Higher = higher BMI

Factor Level Details

sex: *female*, *male*
field_study: *medicine*, *pharmacy*
brightness_adjustment: *on* (enabled), *off* (disabled)
use_reason: *leisure* (social media, entertainment), *reading* (articles, PDFs, books)
chronotype: *morning*, *intermediate*, *evening*
room_lights_off, **device_switched_off**, **device_under_pillow**, **sleep_interrupt_check**, **bedtime_device_use**: *0 = no*, *1 = yes*

Notes

• **Directionality:** PSQI uses “higher = worse” convention; poor sleepers are flagged by **psqi_poor** (>5).
• **Construct validity:** Longer bedtime device use and checking on awakening trend toward worse sleep (higher PSQI, longer latency, lower efficiency). Protective behaviors (e.g., switching device off, reading) trend better.
• **Rounding:** Integers for age and weekly frequencies; decimal precision retained where appropriate (minutes, efficiency %, BMI).
• **Caveat:** This is simulated teaching data; relationships are realistic but not from human subjects.